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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

Experiment – 1

AIM: To create an IAM user with an access key in AWS and to set up a zero-spend budget so that an email alert is sent when AWS spending is incurred.

Description:

AWS Identity and Access Management (IAM) is used to create users and control what each user is allowed to do in an AWS account. An access key (access key ID and secret access key) lets a user work with AWS programmatically, for example through the AWS CLI.

AWS Budgets is a billing service that tracks spending against a limit. A zero-spend budget notifies the account owner by email as soon as spending exceeds $0.01, which helps to stay within the Free Tier.

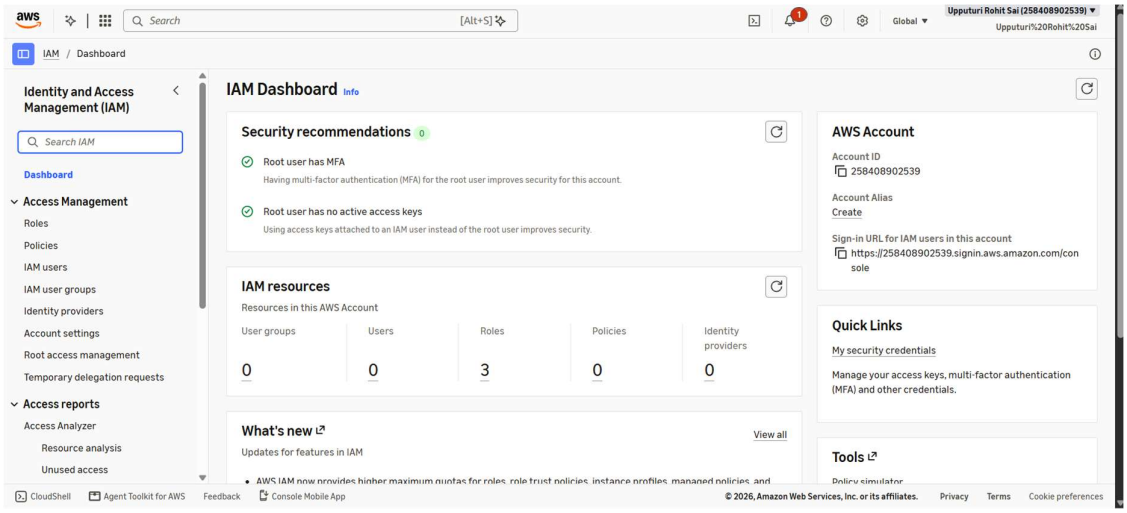
Steps performed:

1. Open the IAM dashboard in the AWS Management Console and select Users → Create user.
2. Enter the user name Rohitsharma@45 and continue without providing console access at this stage.
3. Review the details and click Create user.
4. Open the new user, choose Add permissions → Attach policies directly, select the AdministratorAccess policy, review and add it.
5. Create an access key for the user with the Command Line Interface (CLI) use case and view the access key page. The secret access key is not recorded in this record.
6. Enable console access for the user (autogenerated password) and sign in through the IAM user sign-in page using the account ID, user name and password.
7. Confirm that the AWS Management Console opens for the IAM user.
8. Open the delete dialog for the user as a clean-up step.
9. Open Billing and Cost Management → Budgets → Create budget, select Use a template → Zero spend budget.
10. Enter the budget name and the email recipient, and create the budget.

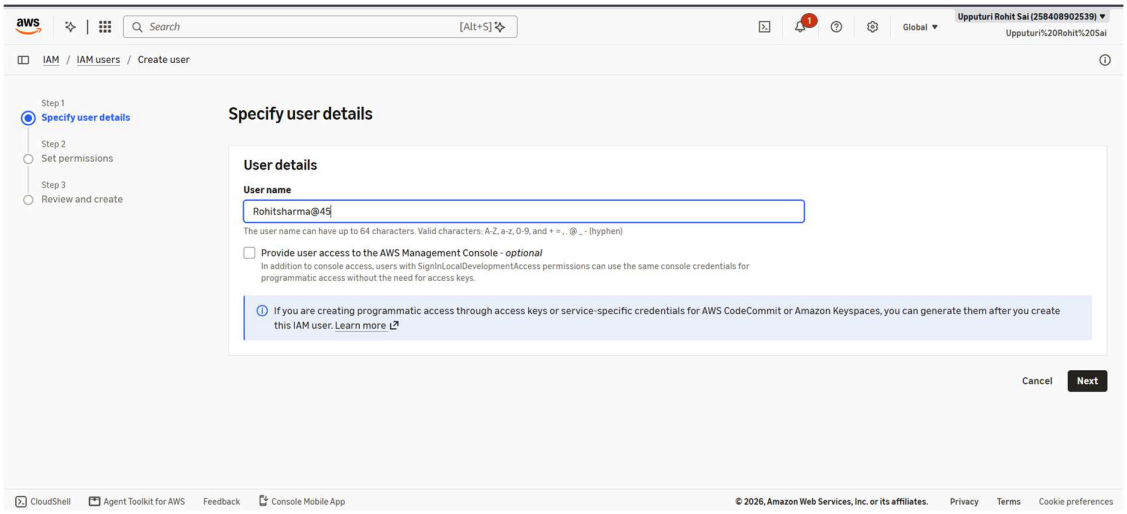
Output:

Creating an IAM user

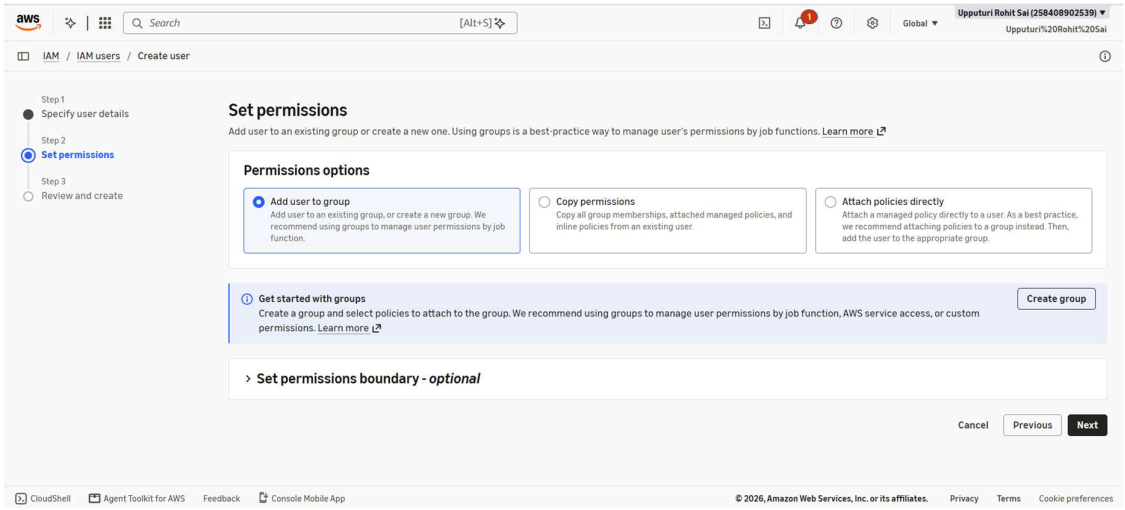
IAM dashboard of the AWS account, showing 0 users and the recommendation "Root user has MFA".



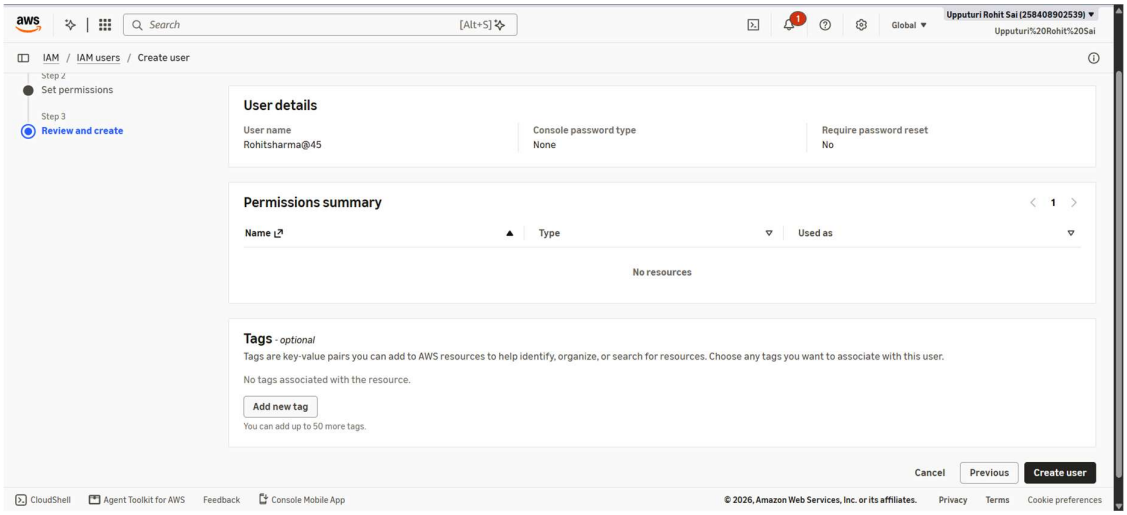
Specify user details: the user name Rohitsharma@45 is entered and console access is left unticked.



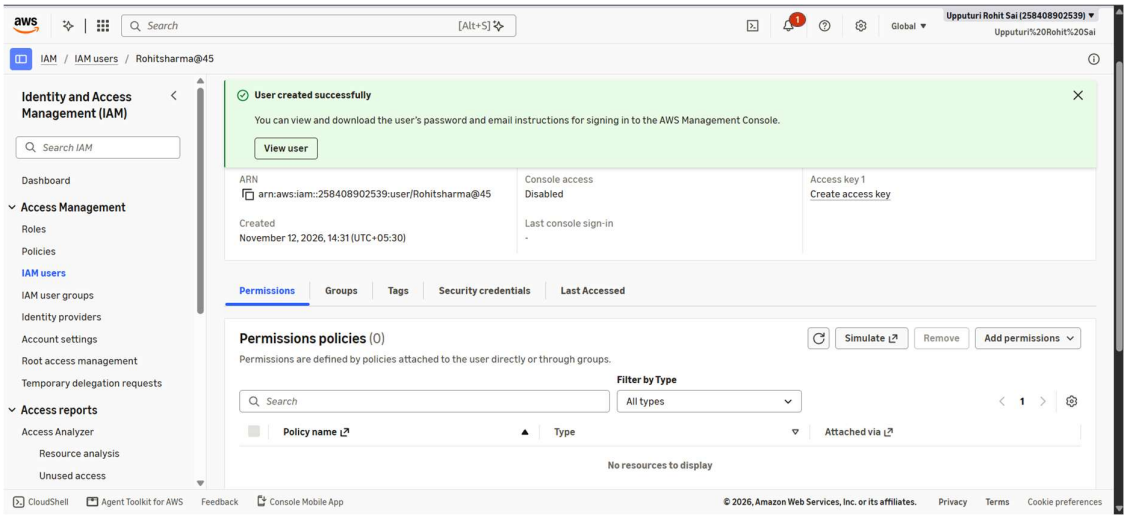
Set permissions step of the Create user wizard, with the Add user to group option displayed.



Review and create: user details and permissions summary before creating the user.

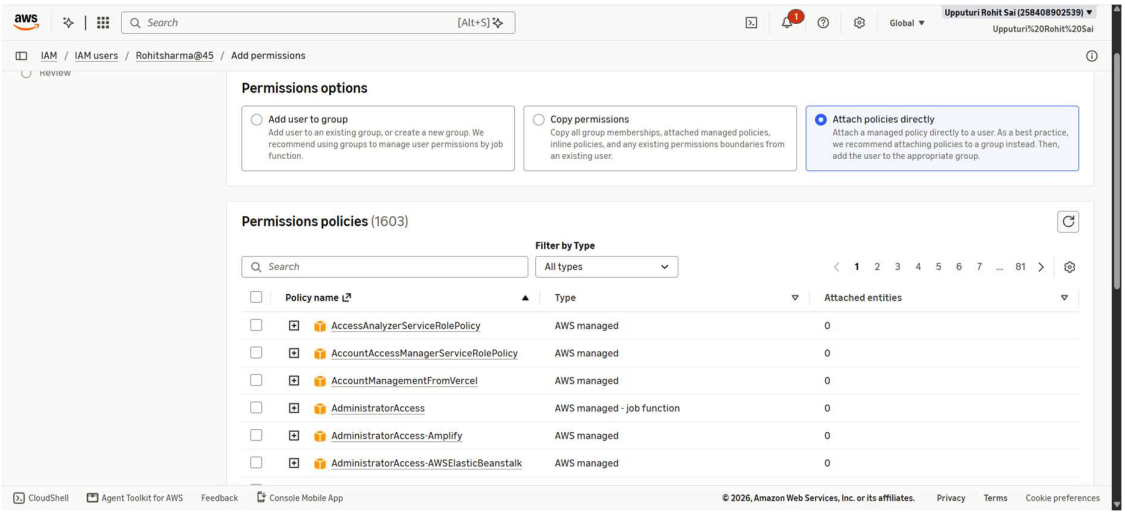


User created successfully. Console access is shown as Disabled and no permission policies are attached yet.

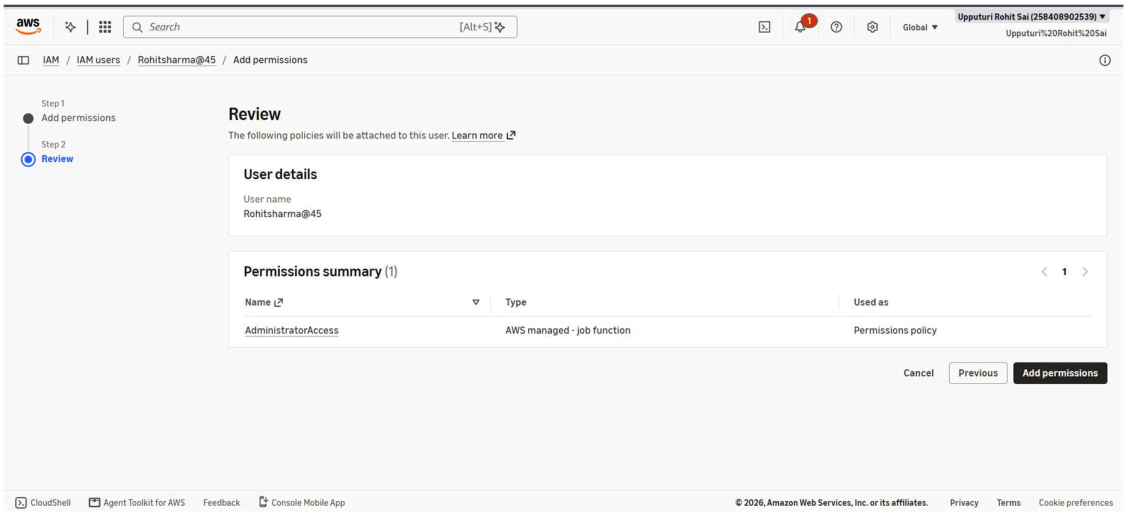


Adding permissions to the user

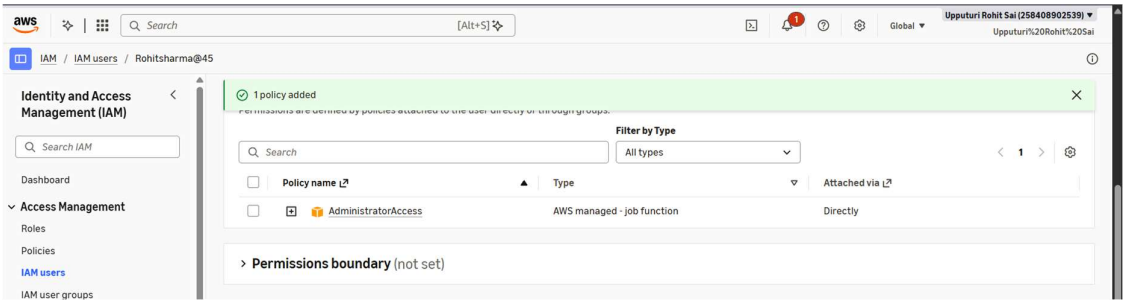
Add permissions for the user: the Attach policies directly option is selected and the AWS managed policies are listed.



Reviewing the policy to be attached to the user: AdministratorAccess (AWS managed - job function).



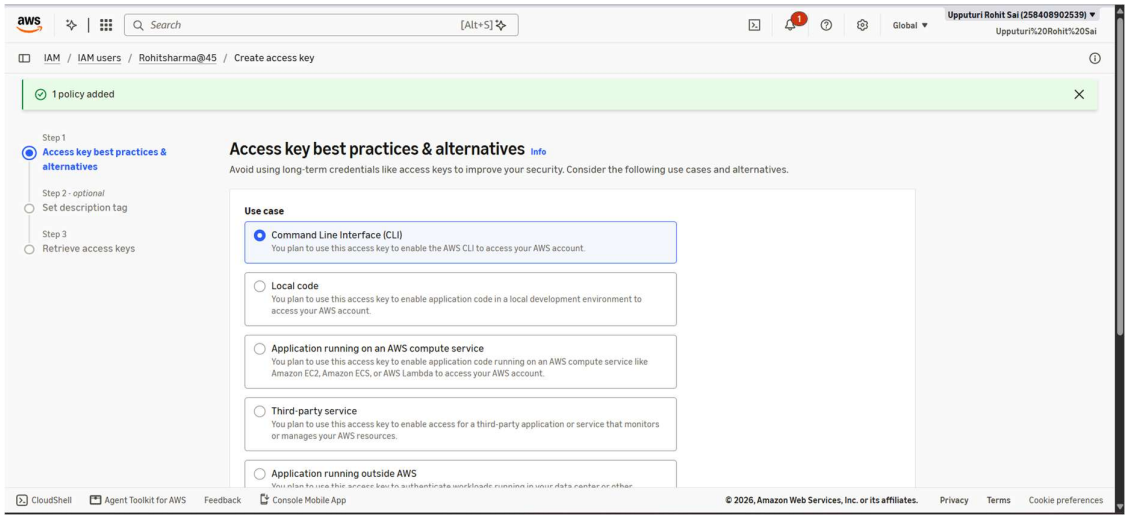
Confirmation that 1 policy was added: AdministratorAccess is attached directly to the user.



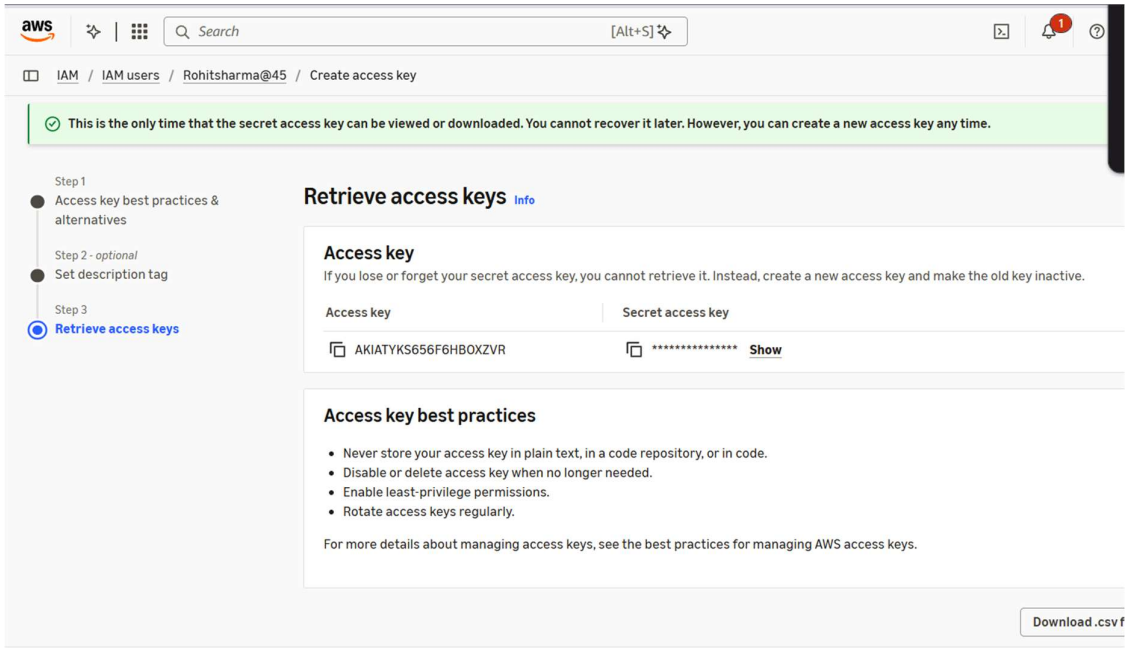
Note: AdministratorAccess gives full access to the account. It is used here for the lab; in real environments, least-privilege policies and groups should be used.

Creating an access key

Creating an access key for the user: the Command Line Interface (CLI) use case is selected on the access key best practices page.

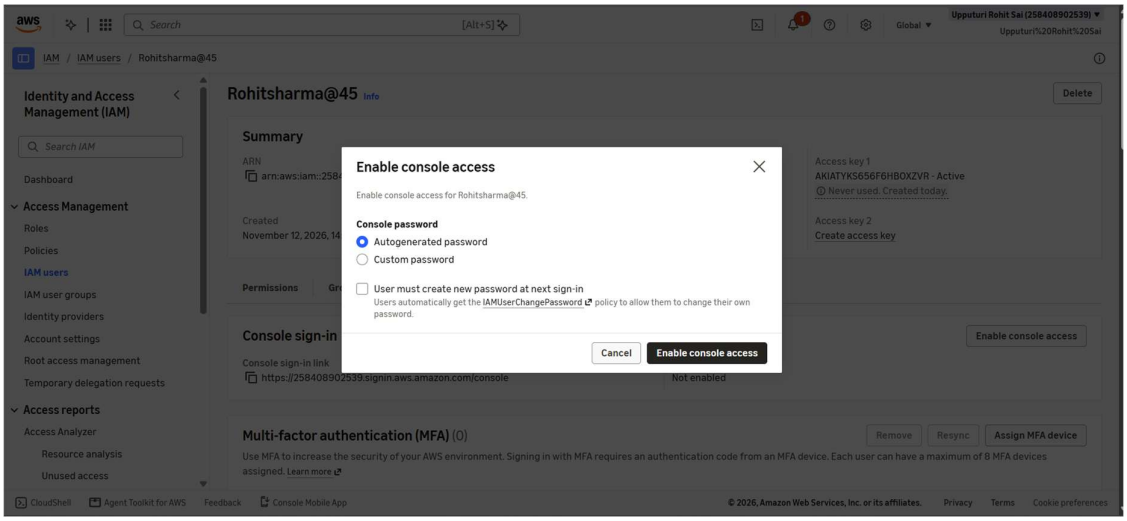


Retrieve access keys page showing the access key ID. The secret access key is masked and is not recorded.

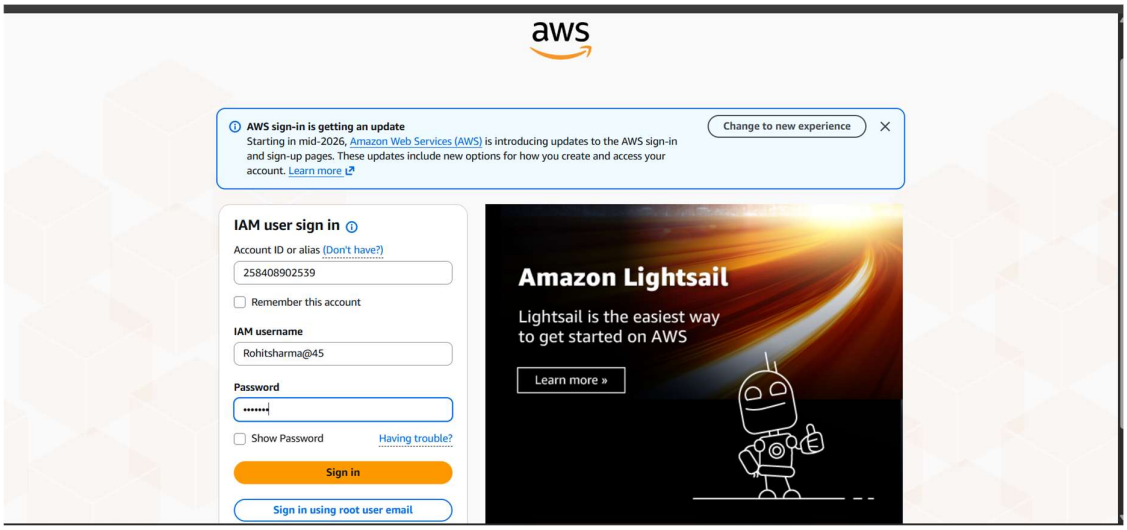


Enabling console access and signing in as the IAM user

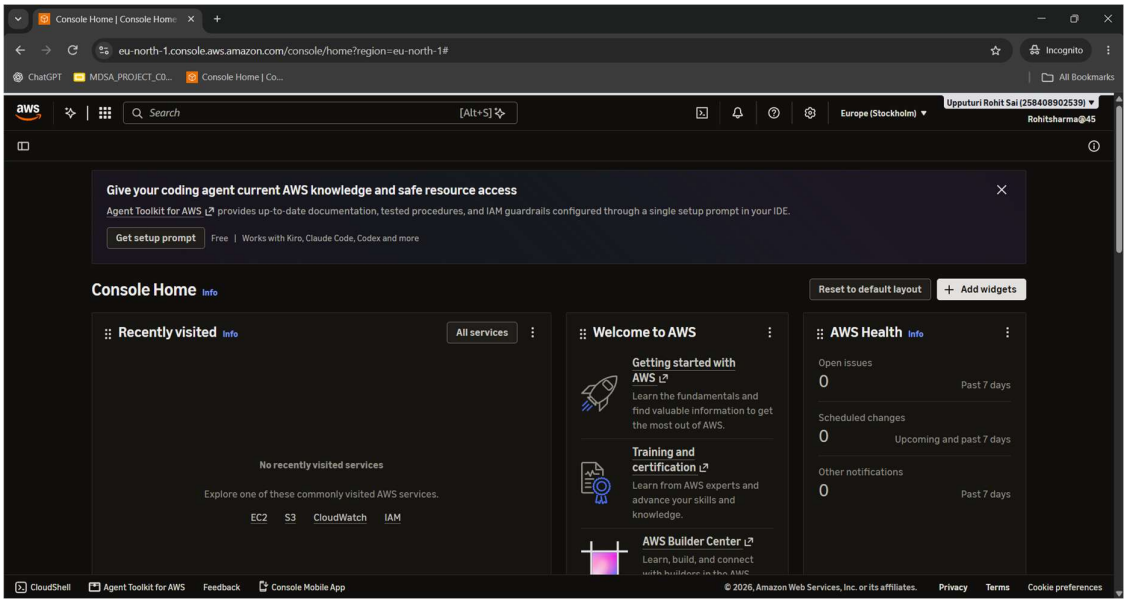
Enable console access dialog for the user, with the autogenerated password option selected.



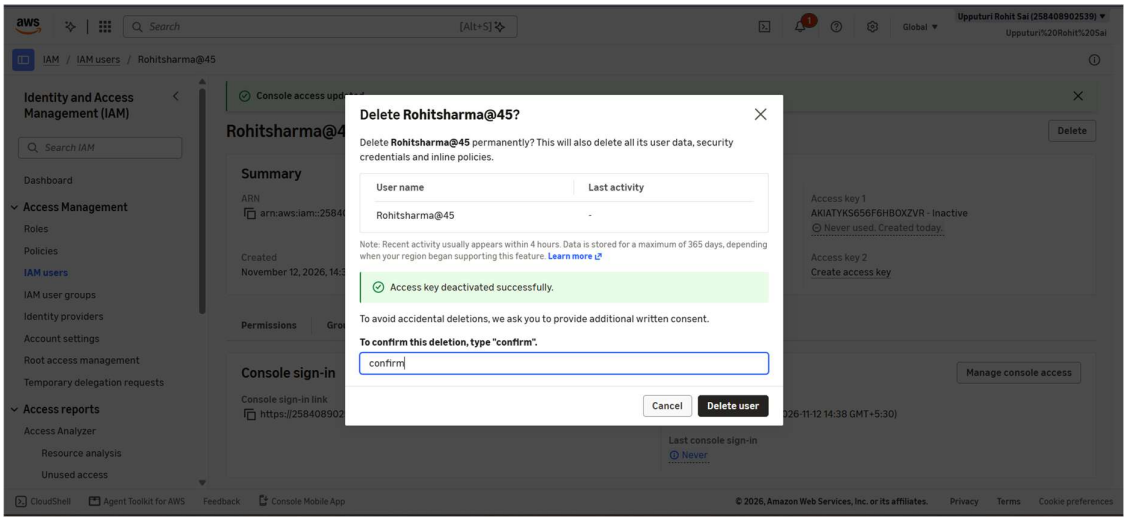
IAM user sign-in page with the account ID and IAM user name entered; the password is masked.



AWS Management Console Home opened in an incognito window, signed in as the IAM user Rohitsharma@45.

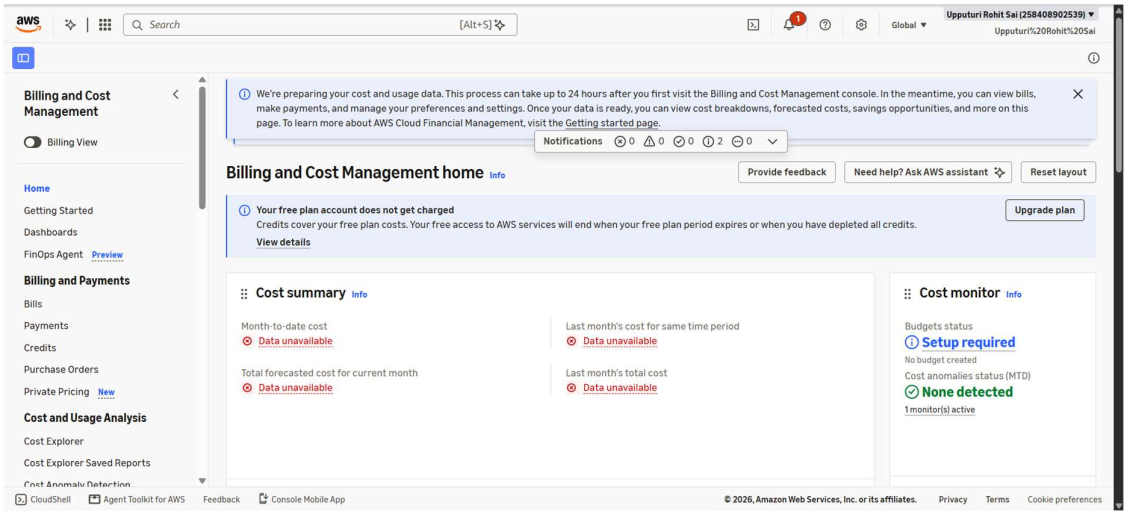


Clean-up: delete user dialog for Rohitsharma@45, showing that the access key was deactivated successfully.

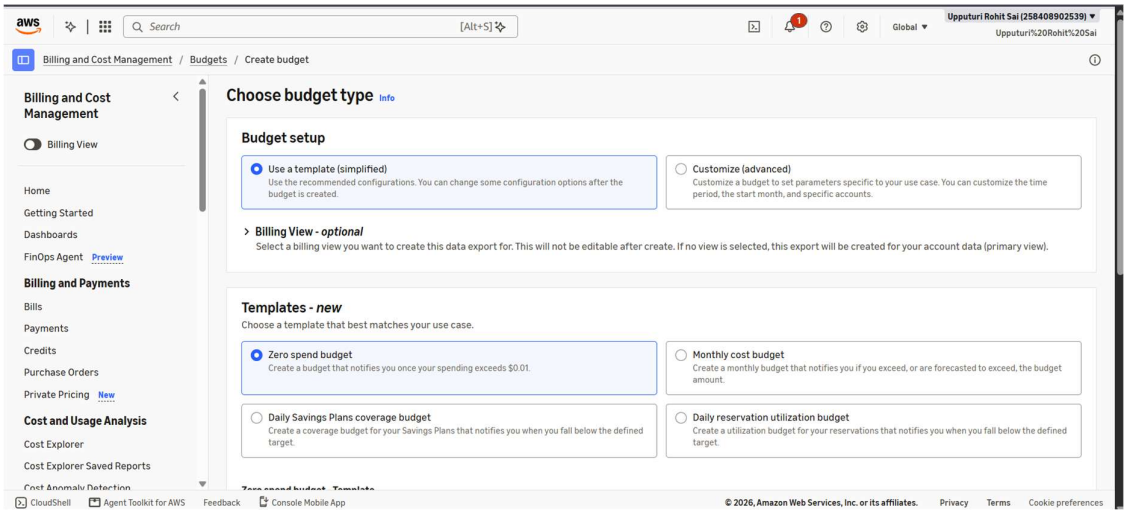


Setting up a zero-spend budget

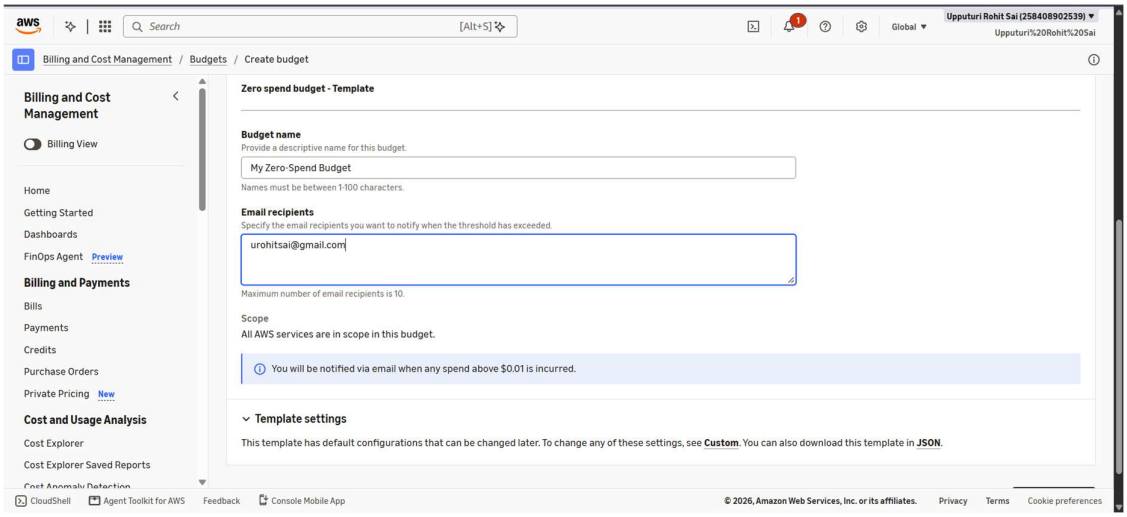
Billing and Cost Management home page of the account.



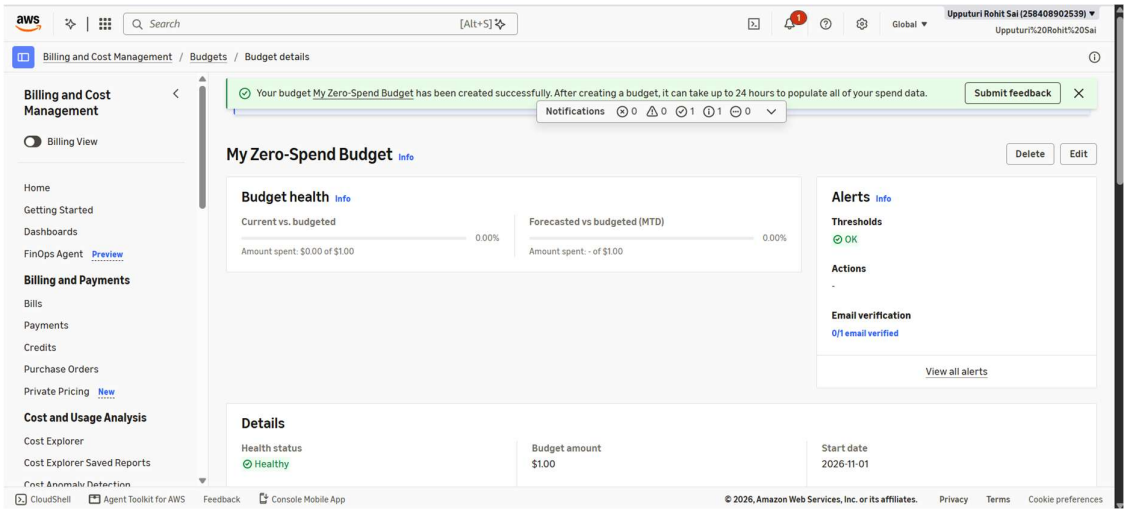
Choose budget type: Use a template (simplified) with the Zero spend budget template selected.



Zero spend budget template: the budget name My Zero-Spend Budget and the email recipient are entered. An email is sent when spend above $0.01 is incurred.



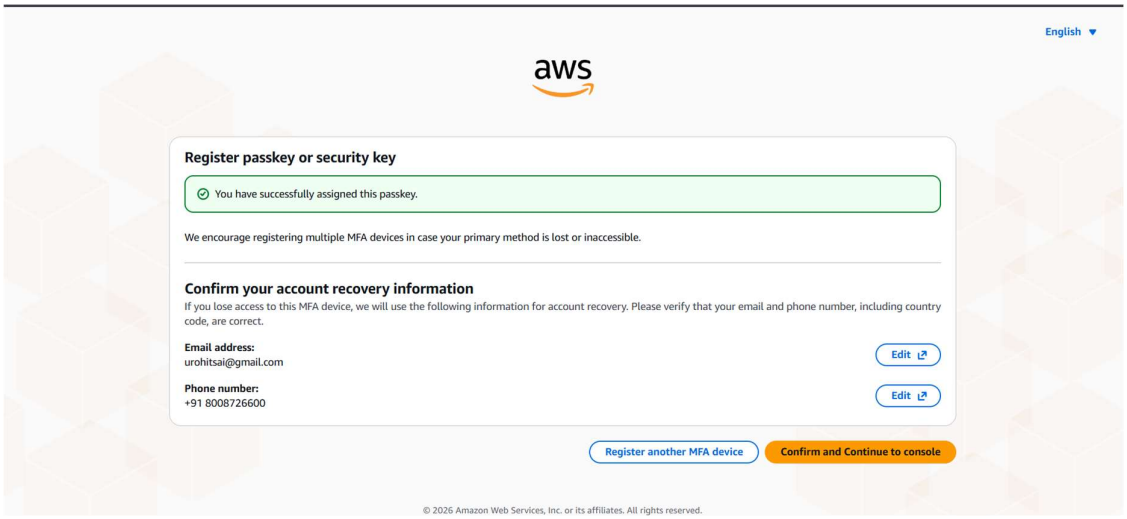
Budget created successfully: My Zero-Spend Budget has a budget amount of $1.00 and the health status Healthy. Email verification shows 0/1 verified at this point.



Practical – session 2

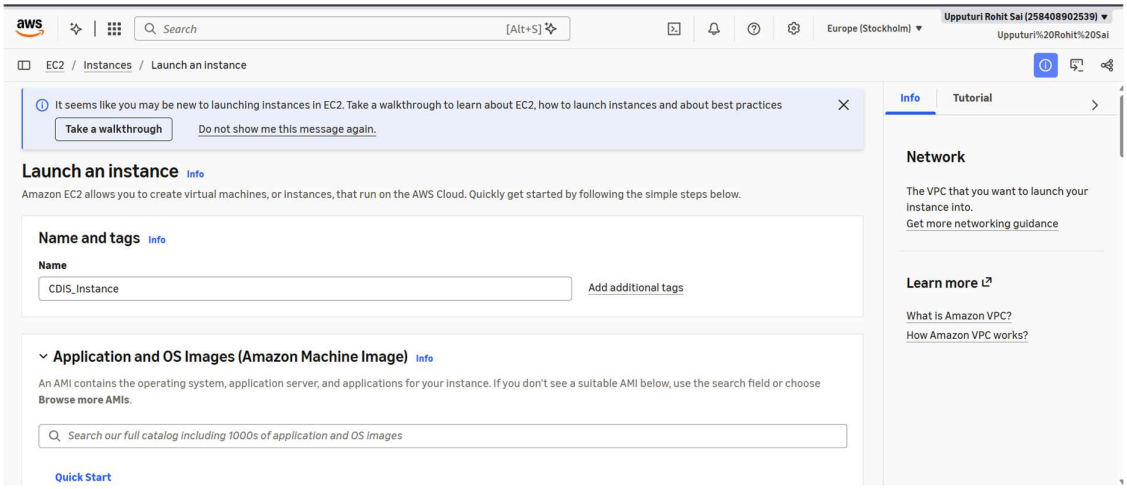
a) Enabling Multi-Factor Authentication to Secure Your Access

Registering a passkey as the MFA device for the AWS account; the console confirms that the passkey was successfully assigned.

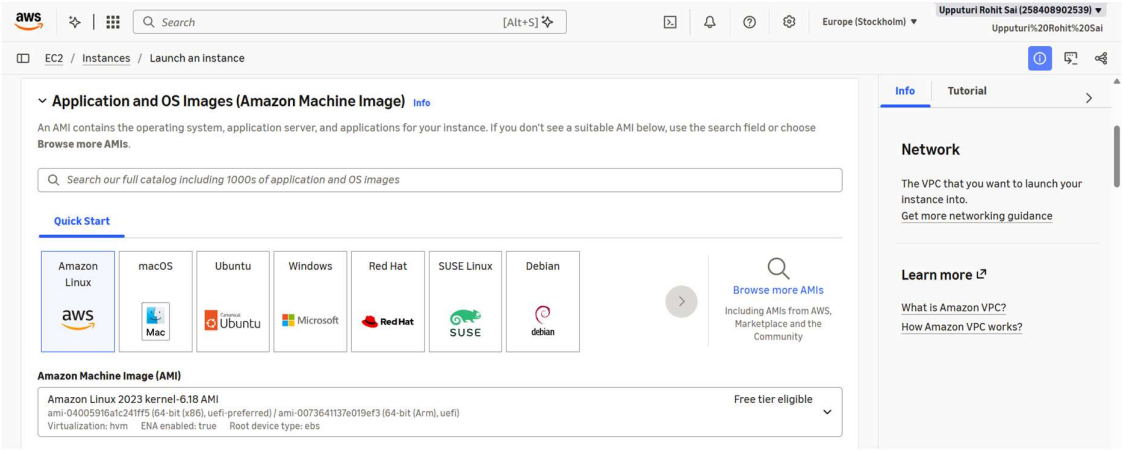


b) Creating First Linux Instance

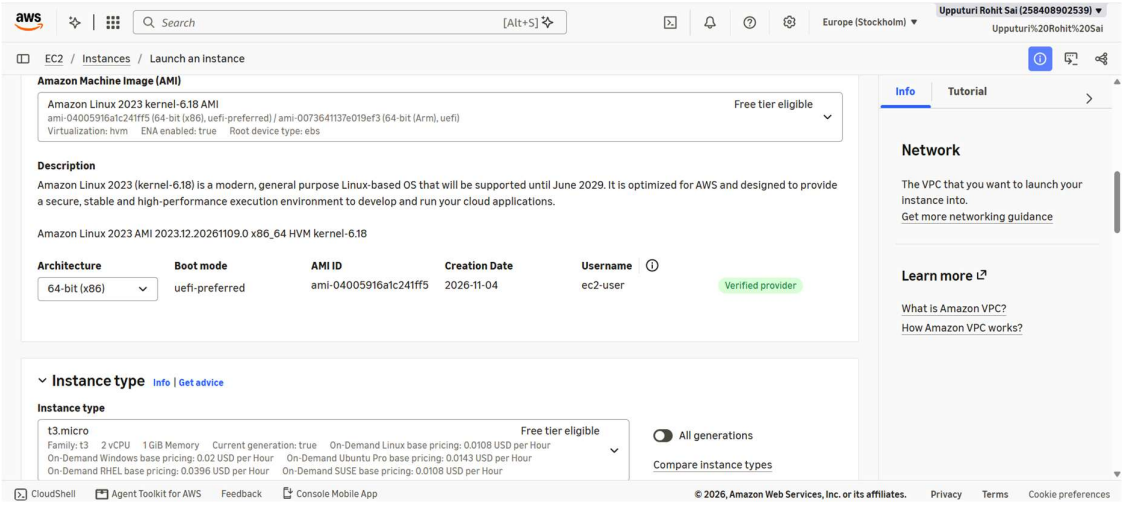
Launching a new EC2 instance by specifying the instance name (CDIS_Instance).



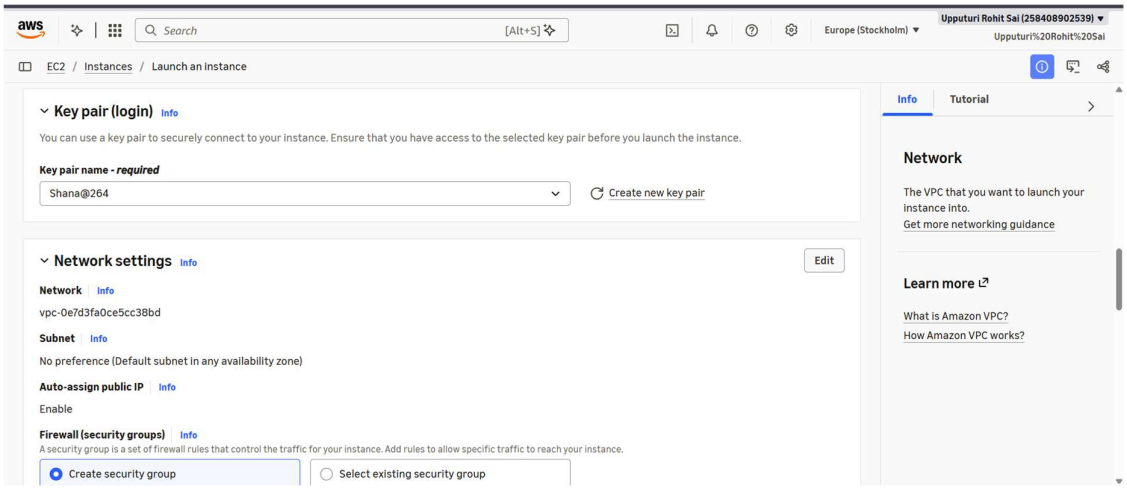
Selecting Amazon Linux in the Quick Start list; the Amazon Linux 2023 AMI (free tier eligible) is chosen.



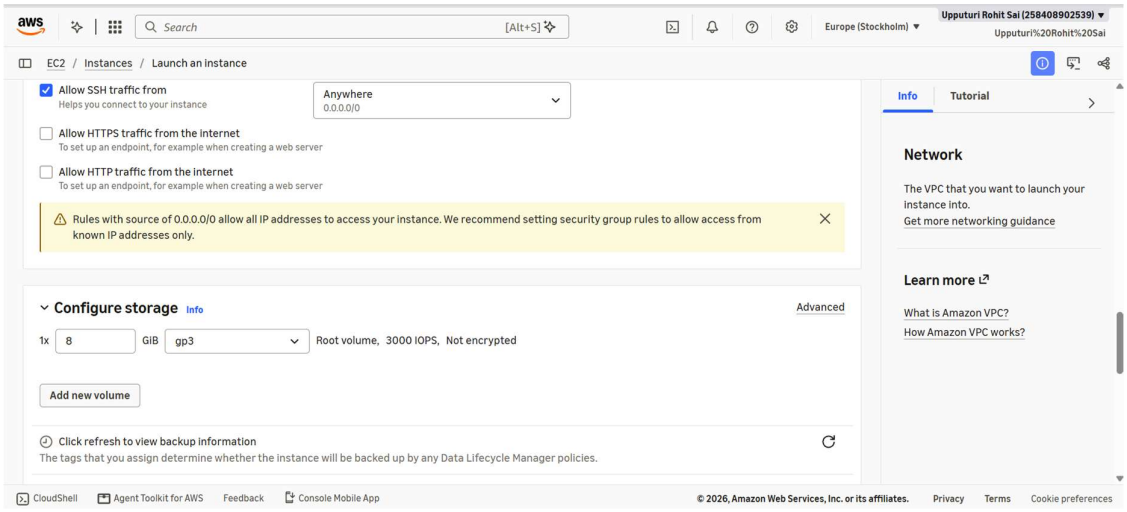
Reviewing the AMI details (64-bit x86, username ec2-user) and selecting the t3.micro (free tier eligible) instance type.



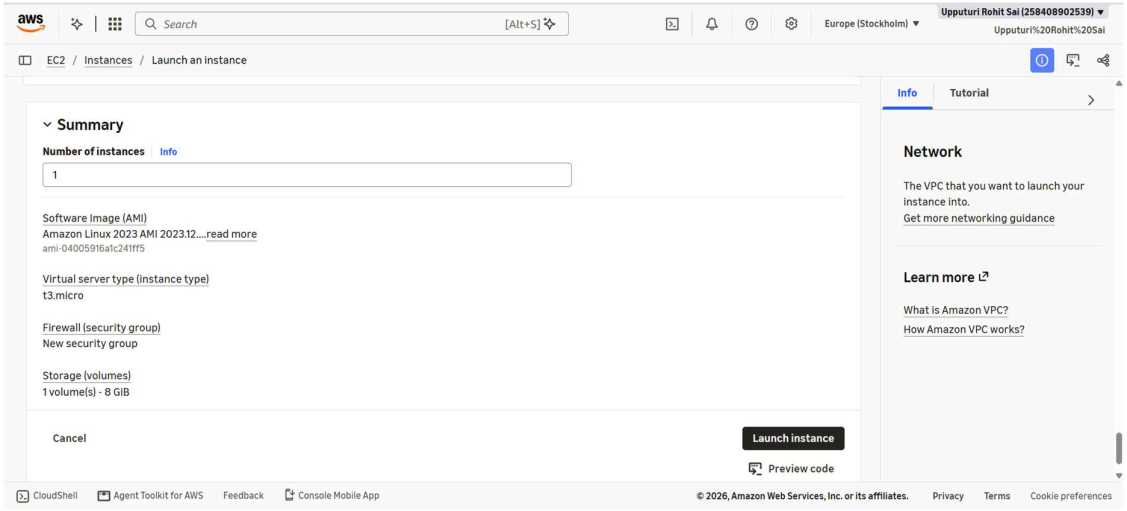
Selecting the key pair and setting up the network configuration: default VPC, auto-assign public IP enabled and a new security group.



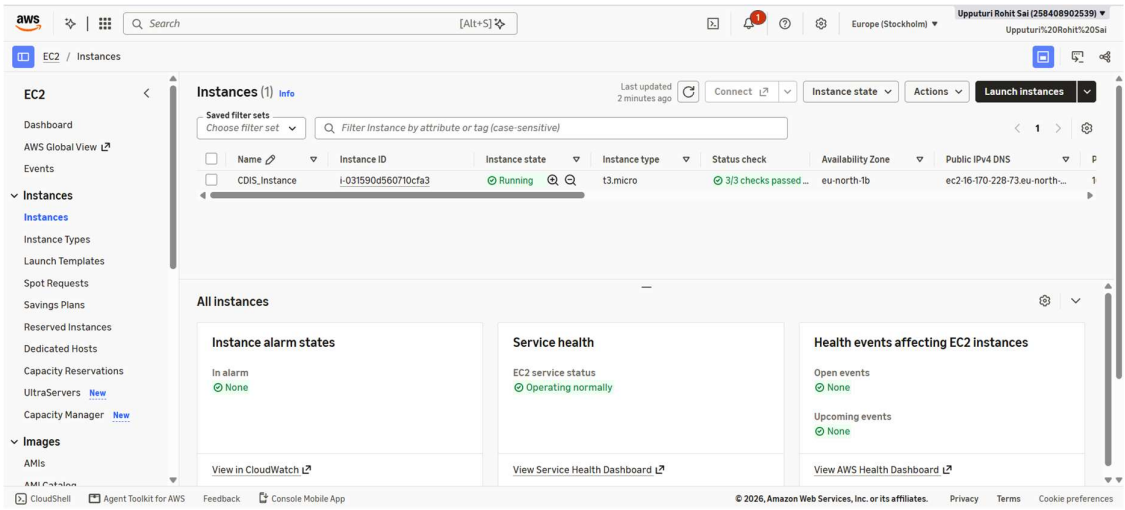
Security group and storage configuration: SSH traffic allowed from anywhere (0.0.0.0/0) and a 1x 8 GiB gp3 root volume.



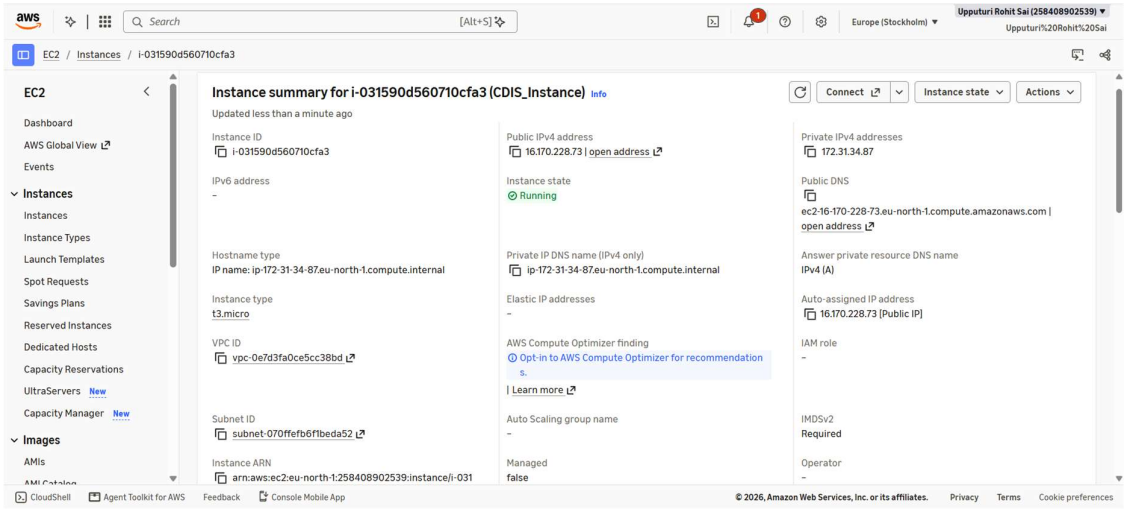
Reviewing the launch summary: Amazon Linux 2023 AMI, t3.micro instance type, new security group and 1 volume of 8 GiB, before launching the instance.



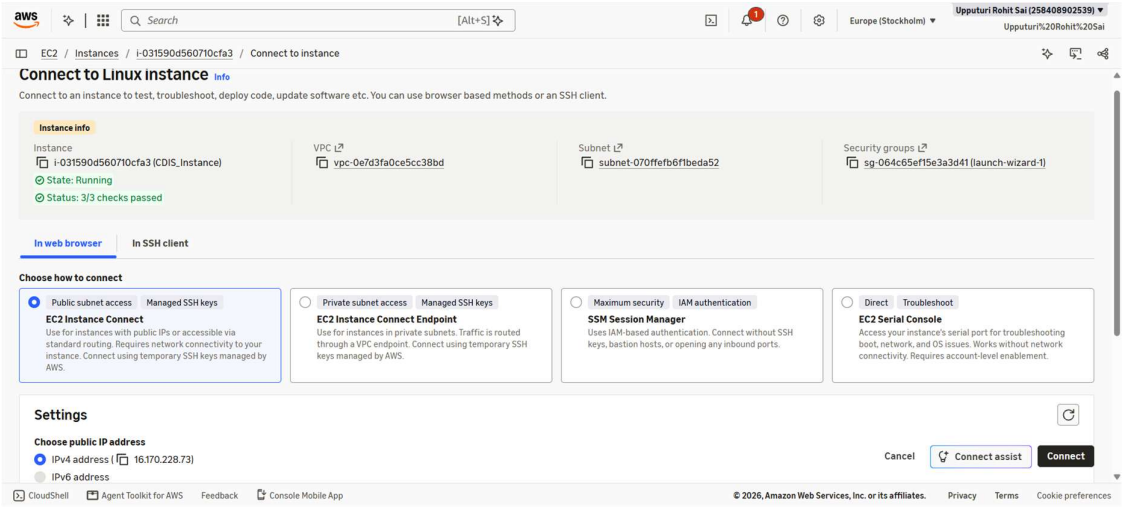
Successful launch of the EC2 instance: CDIS_Instance is in the Running state with 3/3 status checks passed.



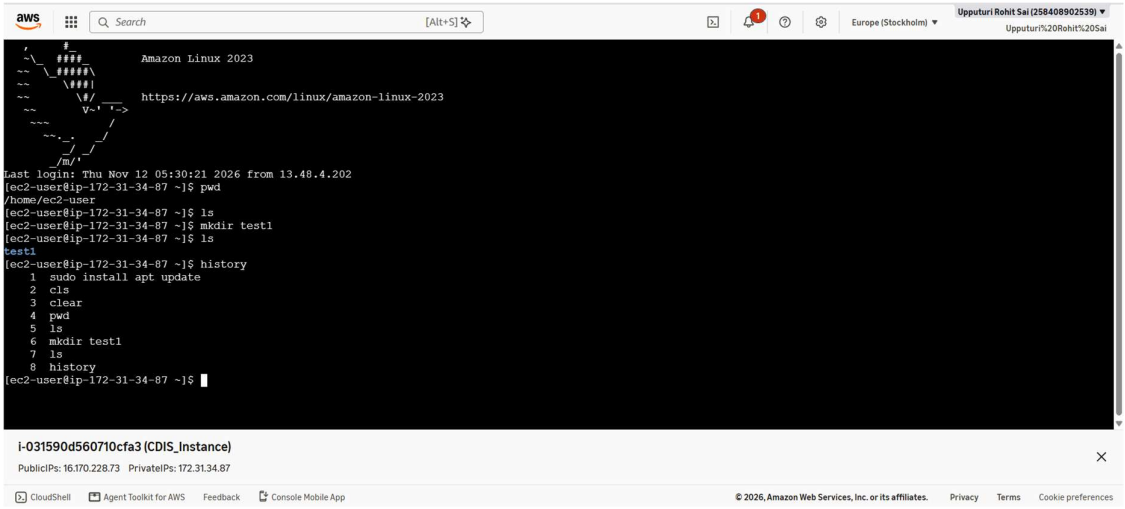
Instance summary showing the instance ID, public and private IPv4 addresses, and the Running state.



Connecting to the instance using EC2 Instance Connect from the web browser, with the public IPv4 address selected.



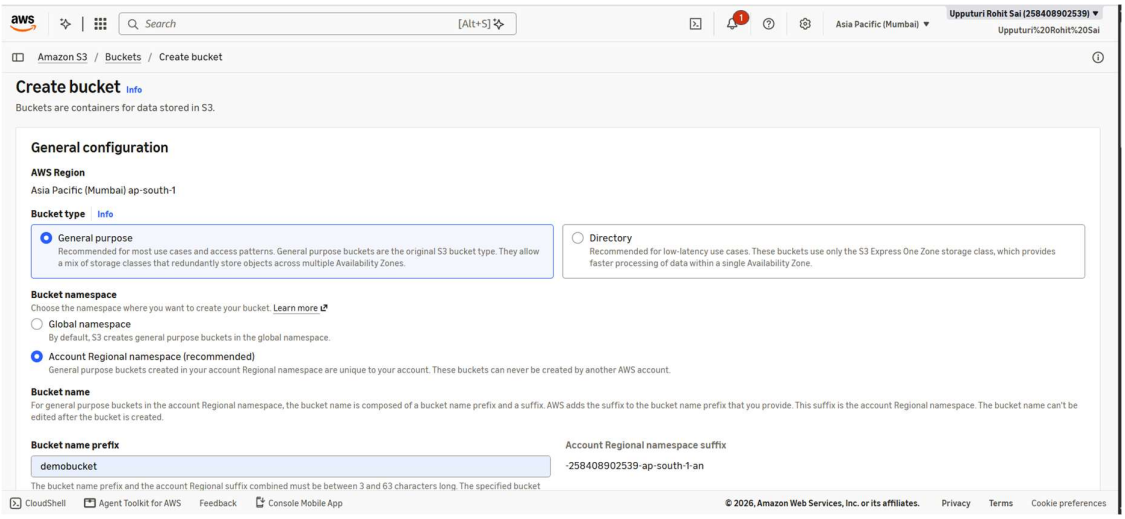
Executing basic Linux commands (pwd, ls, mkdir test1, ls, history) in the terminal to verify access and explore the command-line environment.



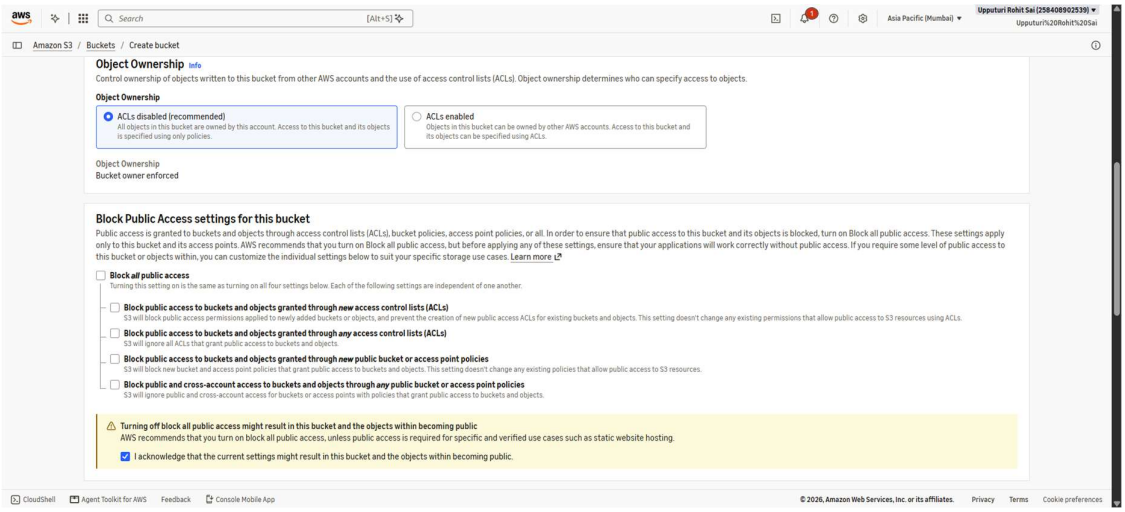
c) S3 Bucket Creation and Upload Objects

I. Create an S3 Bucket for file storage

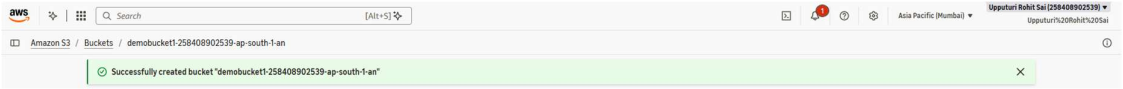
Configuring a new Amazon S3 bucket: general purpose bucket type in the Asia Pacific (Mumbai) region, with the bucket name prefix set to demobucket.



Bucket settings: Object Ownership set to ACLs disabled (bucket owner enforced) and the Block Public Access options turned off, with the acknowledgement checkbox ticked.

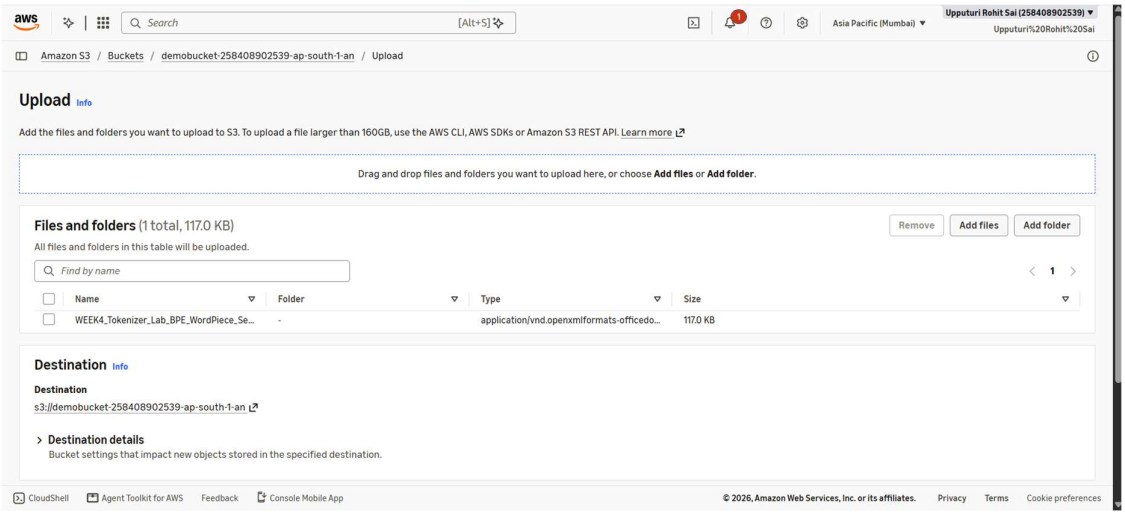


Successful creation of the S3 bucket, confirmed by the success message in the Amazon S3 console.

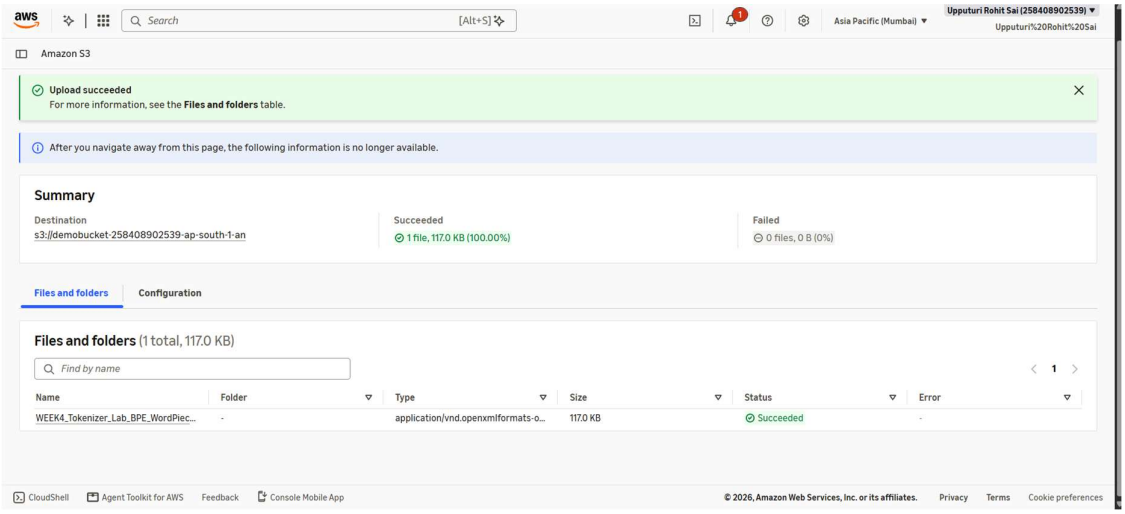


II. Upload objects to the S3 bucket

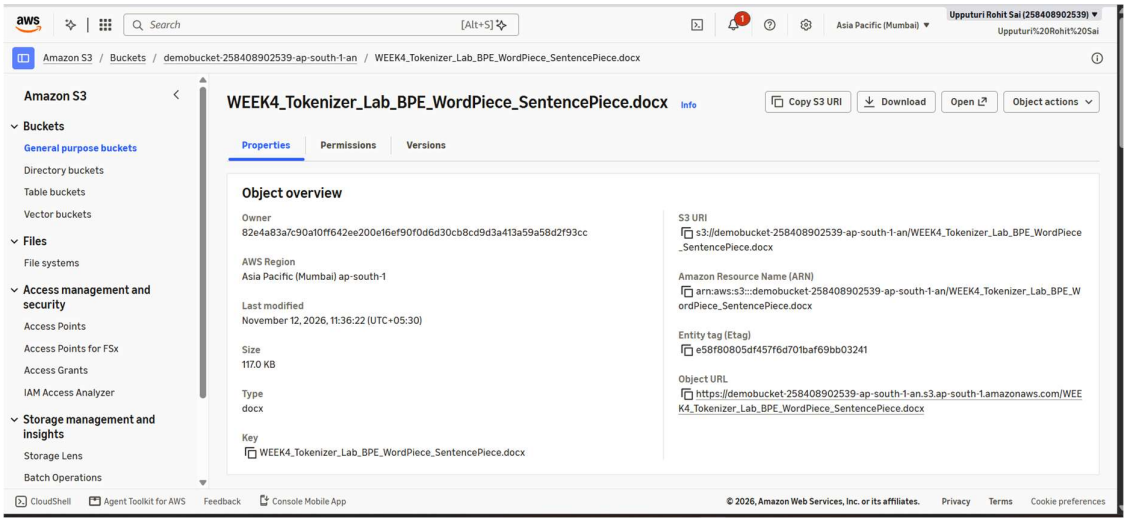
Upload page of bucket demobucket-258408902539-ap-south-1-an with a file selected for upload (.docx, 117.0 KB).



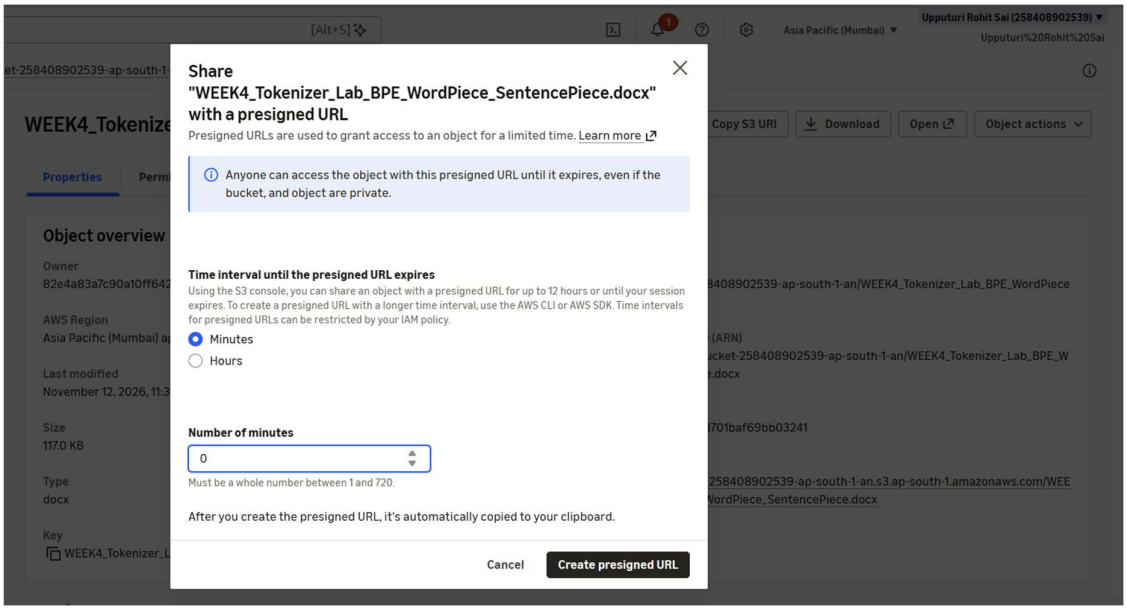
Upload succeeded: 1 file (117.0 KB) uploaded to the bucket, with 0 failures.



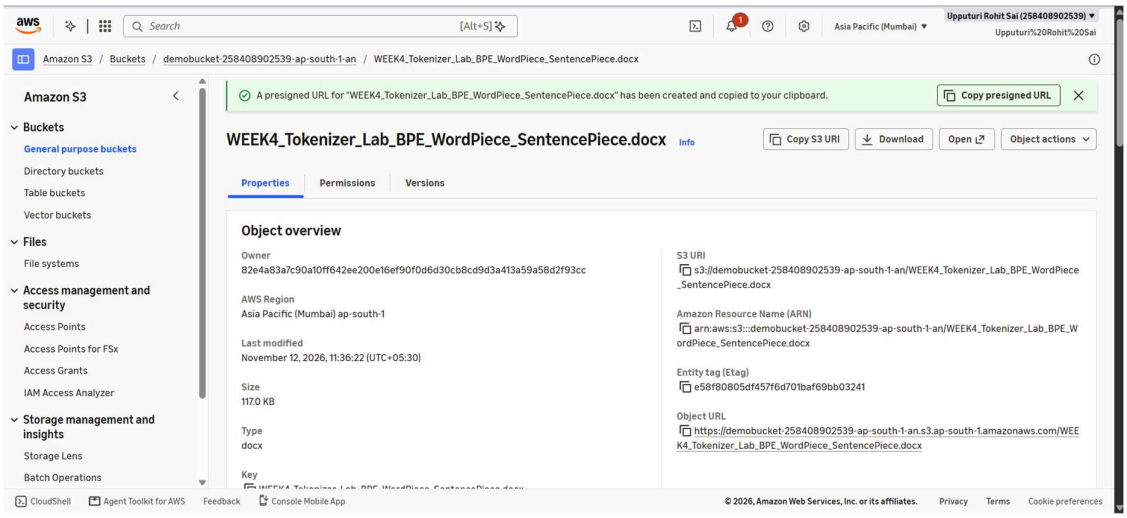
Verifying the uploaded object in the S3 bucket: the .docx object overview shows its S3 URI, ARN and Object URL.



Sharing the uploaded object with a presigned URL, which grants time-limited access to the object.



Confirmation that the presigned URL was created and copied to the clipboard.



Opening the link in an incognito tab: the browser prompts to save the uploaded .docx file, showing that the object is accessible and downloadable.

